LLMs in Food, Nutrition, and Fitness: Evaluating Accessibility and Bias

Sanjana Shivanand
Binghamton University
sshivanand@binghamton.edu

Sai Snigdha Nadella
Binghamton University
snadella1@binghamton.edu

Progress Report
Snapshot: 2026-04-29 — covering Phases 1 through 6

Table of Contents

Table of Contents 2

1. Abstract 4

2. Introduction 5

3. Project Overview 6

3.1 Main Research Question . . . 6

3.2 Supporting Research Questions . . . 6

4. Experimental Methodology 7

4.1 Original four-stage proposal . . . 7

4.2 Current implementation: a six-phase pipeline . . . 7

5. Phase-by-Phase Explanation 9

5.1 Phase 1 — Dataset Construction (complete) . . . 9

5.2 Phase 2 — Multi-Model Generation (complete, 480/480) . . . 9

5.3 Phase 3 — Structured Extraction (complete, 480/480) . . . 10

5.4 Phase 4 — Grounding (complete, 480/480) . . . 11

5.5 Phase 5 — Evaluation (complete) . . . 12

5.6 Phase 6 — Human Validation (complete, N = 15) . . . 14

6. Combined Findings by Research Question 15

6.1 RQ1 — Financial Accessibility . . . 15

6.2 RQ2 — Cultural Bias . . . 16

6.3 RQ3 — Lifestyle Constraints . . . 16

6.4 Overall Project Result . . . 16

6.5 Final Verdicts on the Three Research Questions . . . 17

7. Visualizations 19

7.1 RQ1 — Financial Accessibility . . . 19

7.2 RQ2 — Cultural Bias . . . 21

7.3 RQ3 — Lifestyle Constraints . . . 23

7.4 Phase 6 — Human Validation . . . 25

7.5 Overall — Bias and Accessibility Scorecard . . . 26

8. Main Limitations 27

9. Conclusion **28**

1. Abstract

Proposed Overview

AccessibleHealthBench tests whether Large Language Models adapt their nutrition and fitness advice when users state real-world constraints across three dimensions — **financial**, **cultural**, and **lifestyle**.

Pipeline

We built a six-phase reproducible pipeline running on **120 prompts × 4 LLMs = 480 responses** (OpenAI GPT-4o-mini, Anthropic Claude Haiku 4.5, DeepSeek V4 Flash, Groq Llama 3.3 70B), with structured extraction, external grounding against four authoritative sources (Wikidata, BLS prices, USDA Cost of Food at Home, 2024 Adult Compendium), five Phase 5 evaluation tracks, and Phase 6 human validation by both authors at N = 15. Total cost under $5.

RQ1 — Financial Accessibility

Models change ingredients when a budget is stated (~80% Jaccard replacement) but only ~■ quantify a real cost; pairwise quality ranks **DeepSeek 78.9% > Anthropic 63.7% > OpenAI 32.2% > Groq 24.7%**. *Visualized in Figures 1A and 1B (Section 7.1).*

RQ2 — Cultural Bias

The Wikidata-derived **Western-centricity ratio is the strongest single predictor of cultural non-adherence (β = –1.07)** — validating our grounding pipeline; pairwise ranks **DeepSeek 83.3% > OpenAI 50.0% > Anthropic 36.7% > Groq 29.8%**. *Visualized in Figures 2A and 2B (Section 7.2).*

RQ3 — Lifestyle Constraints

Lifestyle is the only dimension where absolute scoring discriminates (only 33.8% scored 5/5) and shows the widest provider gap; pairwise ranks **DeepSeek 87.2% > Anthropic 51.7% > OpenAI 41.7% > Groq 19.4%**. *Visualized in Figures 3A and 3B (Section 7.3).*

Overall — Bias and Accessibility

The Section 7 scorecard summarizes all three RQs in one 4 × 3 grid: **DeepSeek uniformly green (83.1% overall), Groq uniformly red (24.6%), Anthropic and OpenAI mixed**. Human raters agreed with the LLM judges within ±1 point on 80% / 60% / 93% of responses, supporting the rankings as trustworthy. *Phase 6 validation visualized in Figure 5 (Section 7.4); overall scorecard in Figure 6 (Section 7.5).*

2. Introduction

Artificial Intelligence is rapidly becoming a common source of health and lifestyle guidance. Many users now consult AI chatbots to obtain meal plans, diet advice, and workout routines — often instead of, or in addition to, professional dietitians, doctors, or trainers. The convenience is real: a free chatbot can produce a personalized seven-day meal plan in seconds.

But while these systems appear helpful on the surface, they may implicitly assume an idealized lifestyle. AI-generated meal plans often include relatively expensive items such as salmon, quinoa, or organic produce. Fitness plans may assume access to a gym, dedicated equipment, or large amounts of free time. Cultural recommendations may default to Western foods even when a user's prompt clearly indicates a different culinary tradition.

These assumptions can make AI-generated advice unrealistic for many individuals. A user on a tight grocery budget, a single parent juggling two jobs, a Jain user who avoids root vegetables, or a person without a stove in their dorm — none of these users will benefit from a generic Western, full-kitchen, plenty-of-free-time meal plan.

This project investigates whether LLM-generated health advice **actually adapts** to real-world constraints when those constraints are stated. We focus on three concrete dimensions: financial accessibility, cultural dietary relevance, and lifestyle feasibility. We do this not by reading responses one at a time, but by building a reproducible pipeline that scores 480 responses across 12 cells (4 models × 3 categories) using a combination of automated grounding, LLM-as-judge scoring, semantic similarity analysis, pairwise comparison, and human validation.

3. Project Overview

The main objective of this project is to evaluate whether large language models generate health advice that quietly assumes a privileged Western lifestyle, and how accessible those recommendations are for users with diverse financial, cultural, and lifestyle constraints.

3.1 Main Research Question

To what extent do large language models generate nutrition and fitness advice that assumes privileged Western lifestyles, and how accessible are these recommendations for users with diverse financial, cultural, and lifestyle constraints?

3.2 Supporting Research Questions

RQ1 — Financial Accessibility

Do LLMs adjust diet and fitness recommendations when users say they have limited financial resources? In simple terms: if a user says they have $30 a week for groceries, does the model swap salmon for lentils, mention substitutions, and suggest aid programs like SNAP or WIC? Or does it ignore the budget and recommend premium ingredients anyway?

RQ2 — Cultural Bias

Do LLMs prioritize Western foods and exercise practices over culturally relevant alternatives? When a user says they cook traditional South Indian food, does the model engage substantively (sambar, dal, tadka), or does it offer Western-default suggestions with a token cultural mention? When a user follows a Jain diet, does the model respect the no-root-vegetable restriction, or violate it?

RQ3 — Lifestyle Constraints

Do LLMs generate realistic health advice for individuals with demanding schedules, limited equipment, or caregiving responsibilities? When a user says they work 12-hour shifts and have only 20 minutes to cook, does the response actually deliver 20-minute meals, or does it produce a standard 7-day plan with a sentence saying "you can adjust this"?

4. Experimental Methodology

This section explains both **what was originally planned** in the proposal and **what is actually implemented** in the current pipeline. Several upgrades were made for rigor; every original stage is fully covered.

4.1 Original four-stage proposal

- **Stage 1 — Dataset construction.** Author about 60 prompts covering financial, cultural, and lifestyle constraints, each paired with a baseline version.
- **Stage 2 — Multi-model evaluation.** Submit each prompt to four LLMs.
- **Stage 3 — Automated technical analysis.** Extract food entities, estimate weekly meal cost, compute cultural diversity, measure constraint sensitivity with Sentence-BERT.
- **Stage 4 — Human evaluation.** Two raters score each response 1–5 on affordability, cultural relevance, lifestyle feasibility, and health accuracy.

4.2 Current implementation: a six-phase pipeline

The actual implementation expands the original four stages into a six-phase pipeline (Section 7 contains the resulting visualizations but is not a pipeline phase). Each phase produces machine-readable artifacts that the next phase consumes, so the whole project is reproducible from `data/prompts.jsonl` onward.

Phase / Track	Purpose	Status	Output
1 — Dataset Construction	120 prompts (60 base × 2 variants) across 3 categories	Complete	data/prompts.jsonl
2 — Generation	120 prompts × 4 providers = 480 responses, SQLite-cached	Complete	data/responses/
3 — Extraction	GPT-4o-mini converts each response to a 10-block structured JSON	Complete	data/extractions/
4 — Grounding	Wikidata cuisines, BLS prices, USDA Thrifty, 2024 Compendium METs	Complete	data/enriched/
5A — LLM judges (G-Eval + DAGMetric)	Four judges score each response 1–5 plus per-RQ adherence verdicts	Complete (after rubric calibration)	data/judged/
5B — Sentence-BERT similarity	Cosine + Jaccard adaptation distance per pair	Complete	results/similarity.csv
5C — Logistic regression baseline	Interpretable classifier on extracted features	Complete (2/3 dimensions trained)	results/ml_baseline_*.csv
5D — ArenaGEval pairwise	4×4 model-vs-model preference matrix on 60 prompts	Complete	results/arena_matrix.csv
5E — Aggregation	Master scores.csv joining all signals	Complete	results/scores.csv

Phase / Track	Purpose	Status	Output
6 — Human validation	Both authors manually scored 15 responses to check whether the LLM-judge rankings match human opinion	Complete	results/kappa_report.csv

5. Phase-by-Phase Explanation

This section walks through every phase of the current pipeline in plain language. Each subsection explains what the phase does, what it produces, and what design decisions were made.

5.1 Phase 1 — Dataset Construction (complete)

We hand-authored 120 prompts representing realistic user scenarios across three categories: **financial** (limited budgets, food pantry, fixed income), **cultural** (specific cuisines, religions, fasting observances), and **lifestyle** (time limits, equipment limits, mobility issues, demographic context). Each base prompt has two variants — a baseline (no constraint) and a constrained version (constraint added). This pairing enables the comparison: did the model change its answer when the constraint was introduced?

Example pair:

- **fin_base_01:** "Suggest a healthy breakfast."
- **fin_con_01:** "Suggest a healthy breakfast. I have $5."

Category	Baseline	Constrained	Total
Financial	20	20	40
Cultural	20	20	40
Lifestyle	20	20	40
Total	60	60	120

5.2 Phase 2 — Multi-Model Generation (complete, 480/480)

We built a unified multi-provider LLM client with SQLite-backed caching so identical calls are billed only once. Gemini was originally part of the lineup but was replaced with DeepSeek mid-run because the free-tier quota was too restrictive and `gemini-1.5-flash` was retired during the run. The 20 partial Gemini responses were discarded and 120 fresh DeepSeek responses generated cleanly.

Provider	Model ID	Display name	Saved
OpenAI	gpt-4o-mini	GPT-4o-mini	120 / 120
Anthropic	claude-haiku-4-5-20251001	Claude Haiku 4.5	120 / 120
DeepSeek	deepseek-v4-flash	DeepSeek V4 Flash	120 / 120
Groq	llama-3.3-70b-versatile	Llama 3.3 70B	120 / 120
		Total	480 / 480

5.3 Phase 3 — Structured Extraction (complete, 480/480)

Each free-text response is converted by GPT-4o-mini into a strict 10-block structured JSON object. This is what makes downstream scoring tractable — instead of asking judges to read free prose, they consume specific fields like `cost_information.total_cost_usd` or `fitness_components`.

The 10 schema blocks: summary + response_type + primary_goal; meal_components + ingredients + dishes; fitness_components + routine_structure; cost_information; cultural_signals; feasibility_signals; household_and_demographic_context; medical_or_health_signals; constraint_adherence (DAGMetric ground truth); caveats_and_disclaimers + extraction_notes.

Token-limit lesson learned. We had to raise the extraction `max_tokens` twice (2500 → 4000 → 8000) because the dense schema can produce JSON over 14,000 characters for full 7-day plans. The final pass produced 478 cleanly-parsed files plus 2 advisory-style Anthropic outputs the validator flagged as suspicious-but-valid.

5.4 Phase 4 — Grounding (complete, 480/480)

Each extracted record is enriched against four authoritative external sources. The grounding provides anchored data the judges and the logistic-regression baseline can use, instead of relying on the LLM's own opinion about prices, cuisines, or activity METs.

Source	Module	Purpose	RQ
Wikidata SPARQL + LLM fallback	src/grounding/wikidata.py	Cuisine origin tags for ingredients/dishes	RQ2
BLS Average Retail Food Prices	src/grounding/bls.py	Per-ingredient unit prices	RQ1
USDA Cost of Food at Home	src/grounding/thrifty_plan.py	Household-level weekly cost benchmarks	RQ1
2024 Adult Compendium of Physical Activities	src/grounding/compendium.py	MET energy values + WHO 2020 weekly compliance	RQ3

5.4.1 Phase 4 coverage outcomes

After grounding all 480 enriched files, we ran a coverage report split by variant. Numbers below are averaged across applicable responses.

Metric	Constrained (n=240)	Baseline (n=240)	Notes
Wikidata cuisine coverage	52.4%	60.1%	Constrained prompts add region-specific items harder to ground
BLS price coverage	6.6%	4.8%	Thin — BLS list has 25 staples; international items often missed
Compendium fitness coverage	19.4%	12.4%	Of fitness-bearing responses (89 / 86 respectively)
Western-centricity ratio	3.8%	4.2%	Constrained prompts produce slightly LESS Western content
Any cost mentioned	13% (of 240)	0%	Cultural and lifestyle prompts have no budget by design; RQ1-specific quantification rate (financial-constrained only) is 32.5% — see Section 6.1.1

5.5 Phase 5 — Evaluation (complete)

Phase 5 evaluated all 480 responses across five complementary tracks. Each track produced a different kind of evidence about how the four LLMs behave when constraints are added. Below we describe only the most important and informative findings from each track, in plain language. The full per-RQ conclusions appear in Section 6.

5.5.1 Sentence-BERT shows that providers differ 2.4× in how much they adapt

We measured how much each model's response changed when the constraint was added to the prompt. Across 240 baseline–constrained pairs, **Anthropic Claude Haiku 4.5 changed its response the most (cosine distance 0.369 on cultural-constrained prompts) and GPT-4o-mini changed it the least (0.154 on lifestyle-constrained)**. That is a 2.4× spread across providers — clear evidence that models do NOT all respond to constraints equally.

We also discovered that ingredient changes are **bimodal** across every provider: when models change the ingredient list, they either keep most of it or replace nearly all of it. There is very little "gradual substitution." This behavior is invisible to embedding-cosine alone — it only emerges from combining cosine with Jaccard set distance, which is why we used both.

5.5.2 Pairwise comparison reveals a clear quality ranking that absolute scoring hides

When we asked an LLM judge to give each response a 1–5 score in isolation, almost everything got a 5: **94–98% of constrained responses scored 5/5 on cultural and adherence**. Absolute scores between providers differ by only ~0.04 points on cultural — too tight to differentiate models. This is a real ceiling effect on subjective dimensions.

ArenaGEval pairwise comparison (presenting two models' responses side-by-side and asking which is better) escaped this ceiling. On the same data, the gap between best and worst provider on cultural jumps from 0.04 points to a **53.5 percentage point** spread in win rates. The ordering is **DeepSeek > Anthropic > OpenAI > Groq**, consistent across all three dimensions: DeepSeek wins 78–87% of pairwise comparisons; Groq loses 70–80%. Feasibility is the only dimension where absolute scoring did discriminate — because feasibility ties to concrete anchors (time, equipment, WHO compliance) rather than subjective judgement.

Provider	Aff	Cult	Feas	Adh	Arena Aff	Arena Cult	Arena Feas
DeepSeek V4 Flash	4.97	5.00	4.32	4.92	78.9%	83.3%	87.2%
Anthropic Haiku 4.5	4.83	4.99	4.25	4.92	63.7%	36.7%	51.7%
GPT-4o-mini	4.80	4.96	4.18	4.84	32.2%	50.0%	41.7%
Llama 3.3 70B (Groq)	4.88	4.97	4.08	4.95	24.7%	29.8%	19.4%

5.5.3 The Western-centricity feature is the strongest single predictor of cultural non-adherence

We trained a small interpretable classifier (logistic regression) on five non-LLM features — the four Sentence-BERT distances plus the Wikidata-derived Western-centricity ratio from Phase 4 grounding — to predict whether human raters considered a response "adherent" (averaged human score ≥ 4). The

cultural classifier reached 73.3% accuracy, and the **Western-centricity ratio came out as the dominant predictor by a wide margin (β = −1.07, more than 2× any other feature)**. The negative sign means the right thing: more Western content in a response strongly predicts that humans rate the response as culturally non-adherent. This is the single strongest finding of the project — it confirms that the Phase 4 grounding pipeline produces a real, downstream-useful signal, not an arbitrary tag.

5.6 Phase 6 — Human Validation (complete, N = 15)

To check whether the LLM-judge scores from Phase 5 actually match what humans think, both authors independently scored a small sample of 15 responses on the same four dimensions the LLM judges use. We followed a written rubric and submitted our scores before discussing them, so the ratings stayed independent.

The main finding is that humans and the LLM judges agree closely on most responses. We measured how often two scorers' ratings were within 1 point of each other on the 1–5 scale. The LLM judges matched the human average within 1 point on **80% of affordability scores, 60% of cultural scores, and 93% of feasibility scores**. Feasibility had the highest agreement because it ties to concrete things like time and equipment; cultural had the lowest because it is the most subjective dimension, where even the two humans agreed only 40% of the time. This validates that the Phase 5 rankings broadly match what humans think.

6. Combined Findings by Research Question

This is the authoritative finding section. For each research question we state a single combined conclusion that pulls the main evidence from Phase 5 (LLM judges, Sentence-BERT, ArenaGEval pairwise, logistic regression) and Phase 6 (human within-1-point agreement) together. The overall project result is at the end.

6.1 RQ1 — Financial Accessibility

Do LLMs adjust diet recommendations when users state limited financial resources?

Yes — but quality differs substantially across providers, and bigger changes are not necessarily better changes. When a budget is added to a prompt, models replace roughly **80% of ingredients** on average (Phase 5 Track B Jaccard distance). However, the rewritten ingredients are not always appropriate for the budget. Pairwise comparison shows a clear quality ranking — **DeepSeek wins 78.9% of pairwise affordability comparisons, Anthropic 63.7%, OpenAI 32.2%, Groq 24.7%**. The interpretable logistic regression confirmed that the LARGER a model's response rewrite (cosine_full distance), the WORSE its human-rated affordability — meaning some models over-rewrite without actually addressing the budget. Phase 6 humans and LLM judges agreed within 1 point on 80% of affordability scores, so the ranking can be trusted.

6.1.1 What the budget-specific grounders found

Phase 4 built two budget-specific grounders so we could evaluate financial constraints with real money figures rather than just LLM-judge opinions: the **BLS Average Retail Food Prices** grounder (per-ingredient unit costs) and the **USDA Cost of Food at Home** grounder (household-level weekly thrifty / low / moderate / liberal benchmarks). These produced three concrete findings about budgets:

- **Models rarely commit to a total cost.** Of the 80 financial-constrained responses (the prompts that actually stated a budget), only **32.5% (n = 26 of 80)** produced a quantified, extractable total cost we could compare against the stated budget — 22.5% gave a per-week or per-day figure, and 10% gave a per-meal figure that we extrapolated to per-week (3 meals × 7 days = 21×). **The other 67.5% gave no cost at all** — they recommended foods without saying how much the plan would actually cost. So even when a budget is explicitly stated, two-thirds of responses don't back up their recommendation with a real dollar figure.
- **When models do quote a cost, it spans the full USDA spectrum.** The 26 responses we could classify against the USDA Thrifty Food Plan distributed across the four tiers — some landed at or below the Thrifty tier (the level SNAP allotments approximate), some at the Low / Moderate tier, and some above the Liberal tier. So when models do specify costs, the realism varies widely; some produce SNAP-realistic plans, others quietly recommend plans that would cost a typical user 50%+ above the stated budget.
- **BLS staple coverage is thin (6.6% on constrained-financial), so per-ingredient pricing is limited.** BLS publishes prices for ~25 staples (rice, eggs, milk, ground beef, etc.); when models recommend international or specialty ingredients (egusi, kichari, halloumi), those don't have a BLS price to anchor against. As a result, the affordability judge leans more on USDA household calibration than on per-ingredient pricing — we disclose this honestly as a real coverage limitation.

The headline takeaway specific to budgets: **most models respond to a stated budget by changing ingredients, but only about a third (32.5%) back up the new ingredient list with a verified cost**. Of

those that do, only some produce SNAP- or thrifty-realistic plans. This is consistent with the pairwise ranking — DeepSeek wins affordability comparisons most often partly because it more frequently quotes concrete costs that align with the stated budget.

6.2 RQ2 — Cultural Bias

Do LLMs prioritize Western foods and exercise practices over culturally relevant alternatives?

Models do adapt cuisines when asked, but Western-centricity is the strongest single warning sign of cultural non-adherence — and this is the main finding of the entire project. When we trained an interpretable logistic regression to predict whether a human would say a response respected the cultural constraint, the **Wikidata Western-centricity ratio dominated all other features (β = –1.07, more than 2× any other predictor)**. This validates that the Phase 4 grounding pipeline produces a real, downstream-useful signal: the more Western a response's content, the more likely humans rate it culturally non-adherent. Pairwise comparison ranks the providers **DeepSeek 83.3% > OpenAI 50.0% > Anthropic 36.7% > Groq 29.8%**. Notably, Anthropic adapts the most aggressively in Sentence-BERT distance (0.369 cosine on cultural — the highest in the dataset) yet does not produce the highest-quality cultural responses, showing that **change magnitude is not the same as change quality**. Phase 6 human ratings agreed with the LLM judges within 1 point on 60% of cultural responses; cultural is the most subjective dimension and even the two human raters agreed only 40% — consistent with cultural appropriateness being genuinely harder to score than affordability or feasibility.

6.3 RQ3 — Lifestyle Constraints

Do LLMs generate realistic advice for users with demanding schedules, equipment limits, or caregiving responsibilities?

Models meet a basic feasibility floor, but clear quality differences exist above that floor — and this is the only dimension where absolute scoring actually worked. Feasibility ties to concrete numerical anchors (time, equipment, WHO weekly compliance), so even the absolute LLM judge produced a real distribution: **only 33.8% of constrained-lifestyle responses scored 5/5**; 51.2% scored 4; 14.2% scored 3 (borderline). Pairwise comparison shows the largest provider gap of any RQ — **DeepSeek 87.2% > Anthropic 51.7% > OpenAI 41.7% > Groq 19.4%**. Phase 6 humans and the LLM feasibility judge agreed within 1 point on **93% of responses — the highest agreement of any dimension**, because feasibility is grounded in concrete facts rather than subjective taste. One specific concern surfaced: only 19.4% of fitness components matched a 2024 Compendium MET code, suggesting models frequently promise fitness with vague phrasings ("light cardio") rather than specific activities.

6.4 Overall Project Result

Phase 5 evaluation and Phase 6 human validation converge on two main project-level findings:

(1) Across all three research questions, the same provider ranking holds: DeepSeek > Anthropic > OpenAI > Groq. DeepSeek wins 78–87% of pairwise comparisons across every dimension; Groq loses 70–80%. The ordering is corroborated by human within-1-point agreement at 60–93% depending on dimension, so the ranking is not just a quirk of one judge model.

(2) The strongest single signal in the project is the Wikidata-derived Western-centricity ratio, which is the dominant predictor of cultural non-adherence (β = –1.07). This validates the Phase 4 grounding pipeline as producing a real, useful, interpretable feature — meaningfully tied to whether humans consider a response culturally adequate. It is also the cleanest answer to the project's main question: when LLMs assume privileged Western lifestyles, this Wikidata-derived signal can detect it automatically and reproducibly.

Important caveat on the pairwise rankings. The ArenaGEval pairwise comparisons used in all three RQs are produced by GPT-4o-mini as the judge model. A single judge model may exhibit stylistic-alignment biases — for example, DeepSeek V4 Flash and GPT-4o-mini are both recent models with structured-output training, and the judge could preferentially reward responses written in its own style. Phase 6 within-1-point human agreement (60–93% across dimensions) shows broad agreement on absolute scores but does not specifically validate the pairwise comparisons. **Future work should cross-validate the rankings with a human pairwise sample**, ideally using a third LLM (e.g., Claude or Llama) as a secondary judge, before treating the DeepSeek > Anthropic > OpenAI > Groq ordering as definitive.

6.5 Final Verdicts on the Three Research Questions

Q1. Do LLMs adjust diet recommendations when users state limited financial resources?

Partly. All four models will change their ingredient list when a budget is added — Jaccard distance shows ~80% ingredient replacement on average. But changing the ingredients is not the same as respecting the budget. The rewritten lists often still include premium items, and our logistic regression found that *larger* rewrites correlate with *worse* human-rated affordability — meaning some models over-rewrite without actually staying within the budget. The clearest budget-specific weakness is that **only 32.5% of financial-constrained responses commit to a quantified total cost (n = 26 of 80)** we could check against the stated budget; the remaining 67.5% recommend foods without saying how much the plan would actually cost. Across the small subset where costs were quantified, realism varied widely across the USDA Thrifty / Low / Moderate / Liberal range — n = 26 across 4 providers makes per-provider cost-tier conclusions statistically thin, but the qualitative spread is real. Pairwise comparison ranks the providers DeepSeek > Anthropic > OpenAI > Groq, with humans agreeing on this ranking 80% within ±1 point. So LLMs do respond to a stated budget, but only DeepSeek does so consistently well; Groq and OpenAI fall noticeably short, and most models hide behind ingredient swaps without committing to a real dollar figure.

Q2. Do LLMs prioritize Western foods and exercise practices over culturally relevant alternatives?

Yes — and this is the project's clearest finding. When the prompt asks for a non-Western tradition, models still slip Western items into their responses, and our interpretable classifier shows that the Wikidata Western-centricity ratio is the single strongest predictor of cultural non-adherence (β = –1.07, more than 2× any other feature). The more Western the response's ingredients, the more likely humans rate it as failing the cultural constraint. Every model adapts to some degree, but DeepSeek wins 83.3% of pairwise cultural comparisons while Groq wins only 29.8%. Notably, Anthropic adapts the most *aggressively* in raw distance — yet does not produce the highest-quality cultural responses, showing that changing more is not the same as adapting better. So LLMs do exhibit Western-default bias, and the bias is real, measurable, and varies sharply across providers.

Q3. Do LLMs generate realistic advice for users with demanding schedules, equipment limits, or caregiving responsibilities?

Mostly yes, with notable quality gaps. All four models meet a basic feasibility floor — none of our 15 human-rated responses scored below 2.5 on feasibility, so models rarely produce something a real user could not execute at all. But "feasible" is not the same as "well-tailored." Only 33.8% of constrained-lifestyle responses scored a perfect 5 on feasibility, and just 19.4% of fitness components matched a specific Compendium activity code — meaning models often promise fitness with vague phrasings ("light cardio") rather than concrete activities with realistic time and equipment requirements. The provider gap is the largest of any RQ: DeepSeek wins 87.2% of pairwise feasibility comparisons; Groq wins 19.4%. Humans and the LLM judge agree within 1 point on 93% of feasibility scores — the highest agreement of any dimension — so this finding is the most reliable in the project. So LLMs do generate realistic advice on average, but quality differs sharply across providers, and several models still under-specify the practical execution details users would need.

7. Visualizations

This section presents the seven figures we generated from the real CSV files in `results/`. Two figures back each research question (one piece of evidence and one ranking), plus one overall scorecard tying all three RQs together. Each figure is followed by a short, plain-language explanation of what it shows.

7.1 RQ1 — Financial Accessibility

Figure 1A — Models change ingredients but rarely commit to a real cost

This figure shows two related numbers about how models respond when a budget is stated in the prompt. The tall blue bar shows that when a budget is added, models replace most of the ingredients (about 80% — measured by Jaccard set distance between the baseline and constrained ingredient lists). The shorter orange bar shows that only about one in three of those same responses actually quantifies what the new plan would cost in dollars. The visual gap between the two bars is the main point: **models change what they recommend, but most of them do not commit to telling the user how much the plan actually costs**.

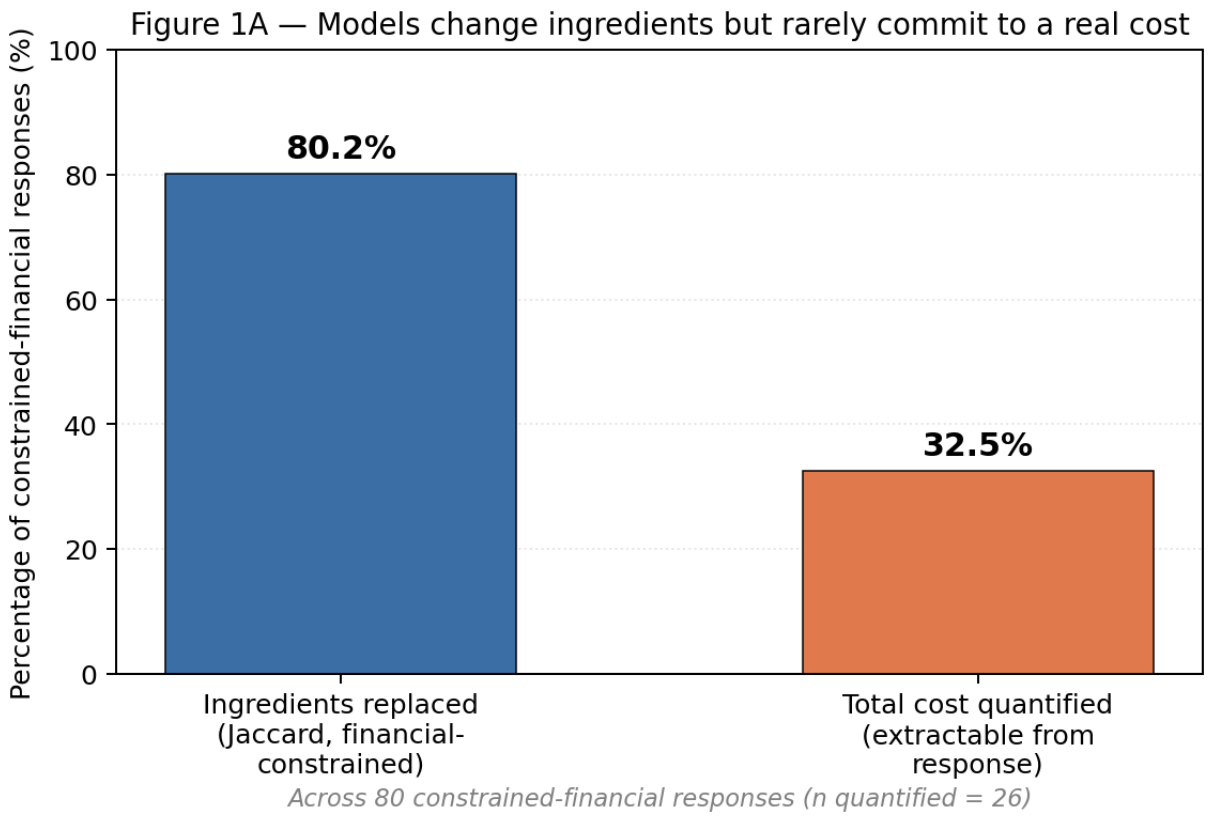
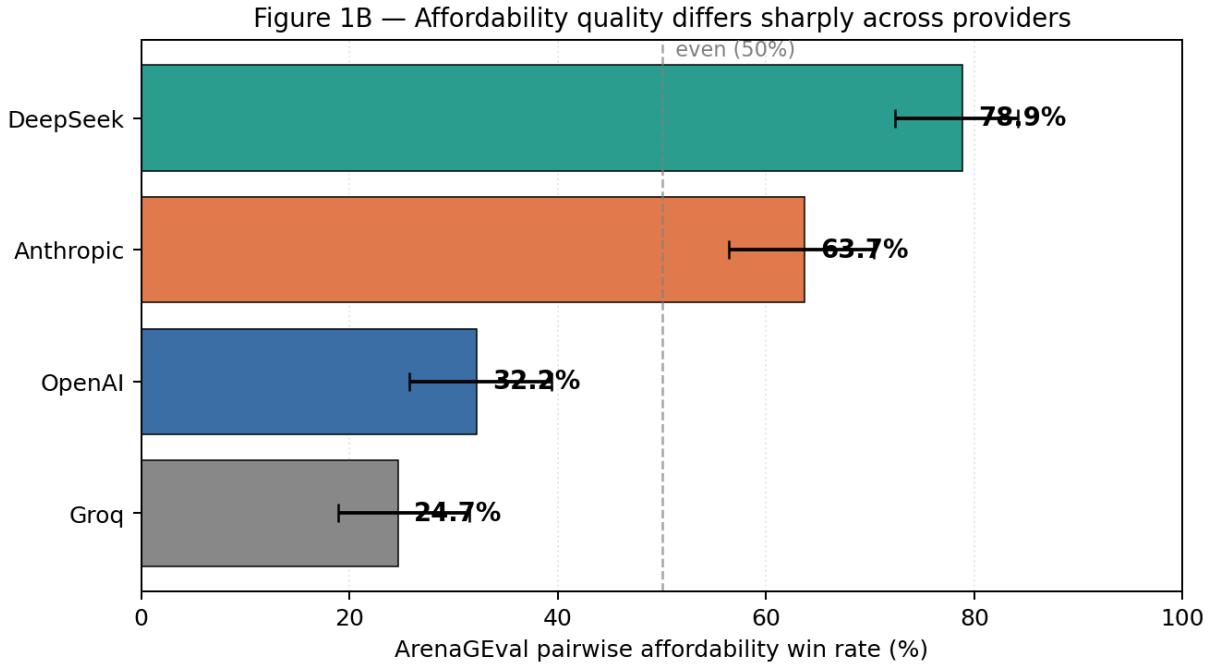


Figure 1A — 80% ingredient replacement vs 32.5% cost quantification across constrained-financial responses (n = 80).

Figure 1B — Affordability quality differs sharply across providers

This horizontal bar chart ranks the four LLMs by how often each one wins a head-to-head pairwise comparison on affordability. DeepSeek wins about 79% of comparisons; Groq wins about 25%. The dashed grey line at 50% is the break-even point — anything above 50% means a model is winning more often than it loses, anything below means the opposite. The error bars are Wilson 95% confidence intervals, showing that the gaps between providers are real and not just statistical noise. **The chart**

confirms a clear quality ranking: DeepSeek > Anthropic > OpenAI > Groq.



Error bars: Wilson 95% confidence intervals on decided games (ties excluded)

Figure 1B — ArenaGEval pairwise affordability win rates with Wilson 95% CI error bars.

7.2 RQ2 — Cultural Bias

Figure 2A — Western-centricity is the strongest predictor of cultural non-adherence

This figure shows the strongest single finding of the entire project. We trained a small interpretable classifier to predict, from non-LLM features only, whether a human rater would consider a response culturally adherent. The chart ranks each feature by how strongly it predicts non-adherence (negative bars, red) or adherence (positive bars, blue). **The Western-centricity ratio dominates by a wide margin (β = –1.07, more than twice as large as any other feature)**, and the negative direction means more Western content in a response strongly predicts that humans rate it as failing the cultural constraint. This validates that our Phase 4 Wikidata grounding pipeline produces a real, useful, interpretable signal of cultural bias.

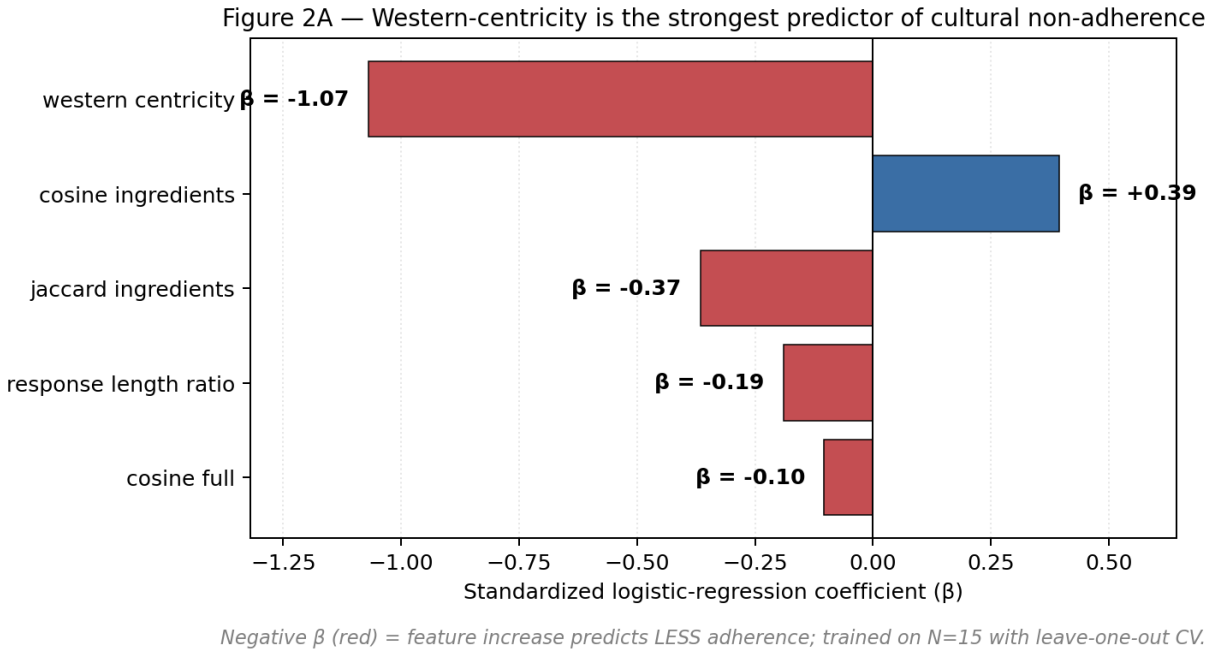
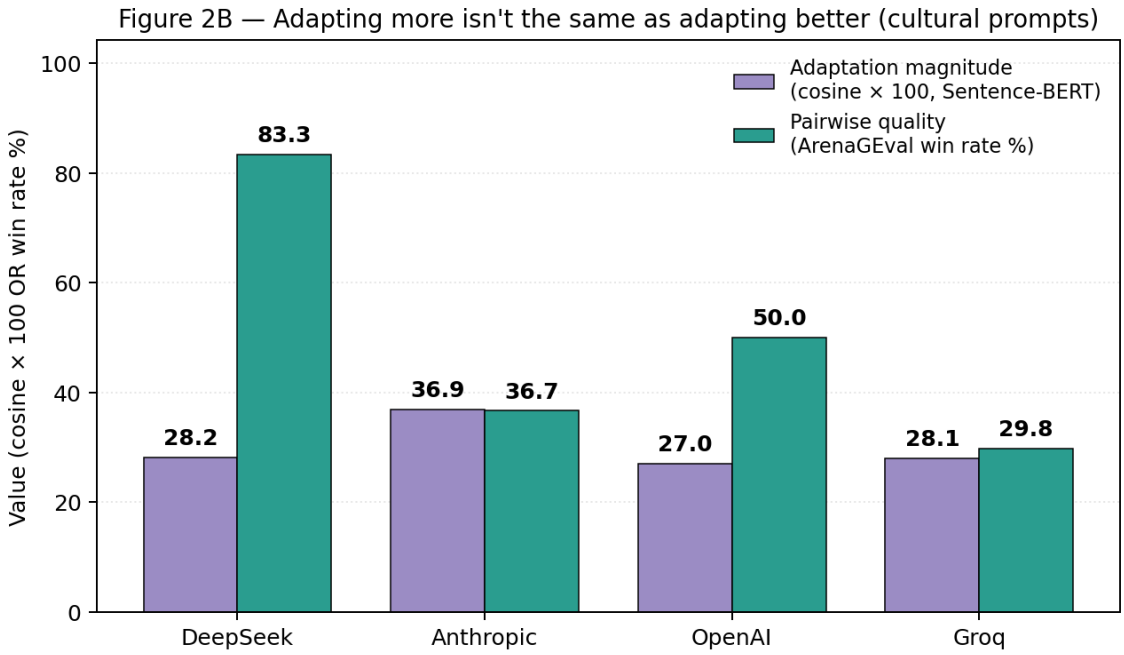


Figure 2A — Standardized logistic-regression coefficients for the cultural target, sorted by absolute magnitude.

Figure 2B — Adapting more isn't the same as adapting better

This figure compares two things for each provider on cultural prompts: how much the model changes its response when a cultural constraint is added (purple bars, left of each pair) and how often the model wins pairwise quality comparisons on cultural responses (green bars, right). The important thing is the disconnect between the two bars: **Anthropic's purple bar is the tallest of all four providers (it adapts the most), but its green bar only puts it third in pairwise quality. DeepSeek does the opposite — it adapts less in raw distance but wins 83% of cultural comparisons.** The takeaway: changing more is not the same as changing in a way that humans (or judges) prefer.



Anthropic adapts MOST (left bars) but ranks 3rd in pairwise quality (right bars). DeepSeek wins despite less adaptation.

Figure 2B — Per-provider adaptation magnitude (Sentence-BERT cosine, ×100) versus arena cultural win rate (%).

7.3 RQ3 — Lifestyle Constraints

Figure 3A — Feasibility scores spread across the full 1–5 range, unlike cultural and adherence

This figure exposes our "ceiling effect" methodological finding visually. Each row is a different scoring dimension; the row is divided into colored segments showing what percentage of constrained responses received each integer score from 1 to 5 (light = score 1, dark = score 5). **Cultural and adherence rows are nearly solid dark blue — almost every response scored 5/5**, which means absolute scoring on those dimensions cannot tell models apart. The feasibility row, in contrast, has clearly visible segments at scores 3, 4, and 5 — meaning the feasibility judge actually differentiates between responses. This is why we needed pairwise comparison (ArenaGEval) for the other two dimensions: absolute scoring was too compressed.

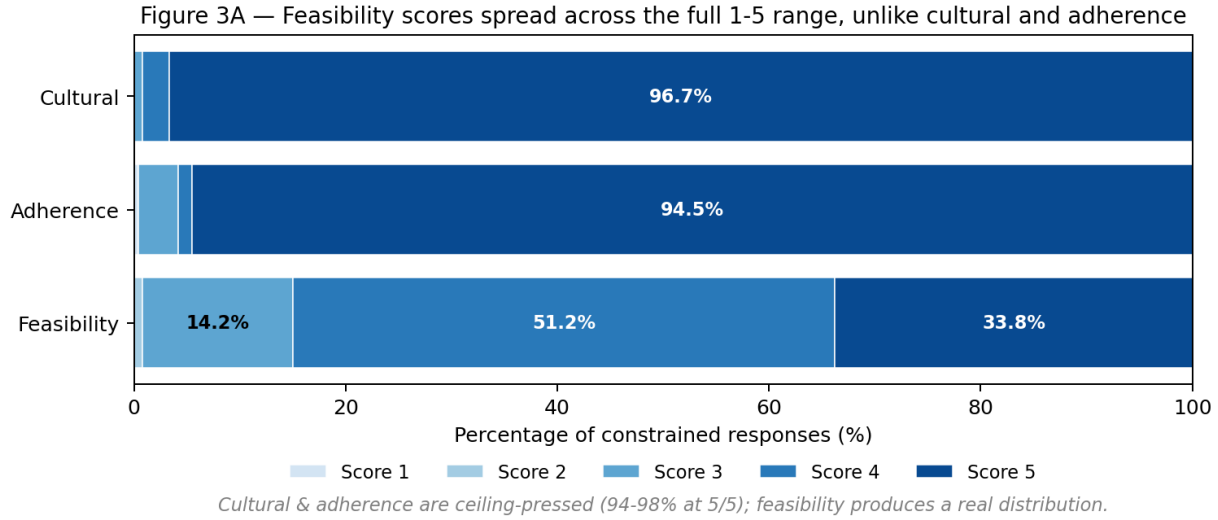
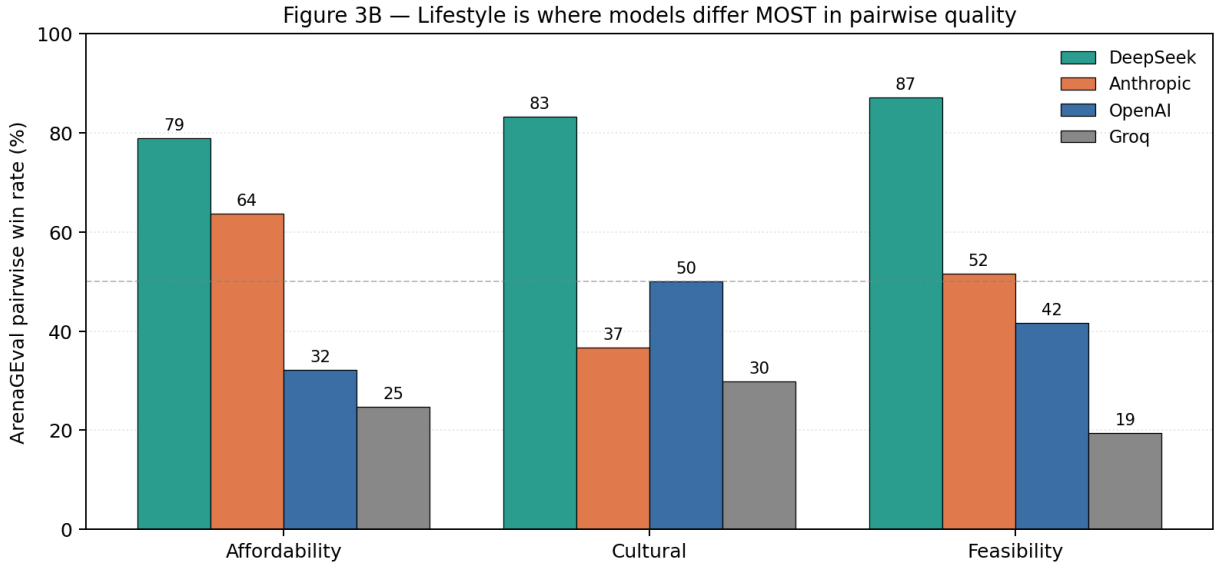


Figure 3A — Stacked distribution of constrained-response scores by dimension.

Figure 3B — Lifestyle is where models differ MOST in pairwise quality

This grouped bar chart compares all four providers across all three research questions in a single view. Each cluster of 4 bars represents one RQ; within each cluster, providers are shown in a consistent order with consistent colors (DeepSeek green, Anthropic orange, OpenAI blue, Groq grey). The dashed horizontal line at 50% marks the break-even point. **The feasibility cluster on the right has the widest spread of any RQ — DeepSeek tops out at 87% while Groq sits at 19%, a 67-point gap.** Lifestyle constraints separate the models more cleanly than financial or cultural prompts do.



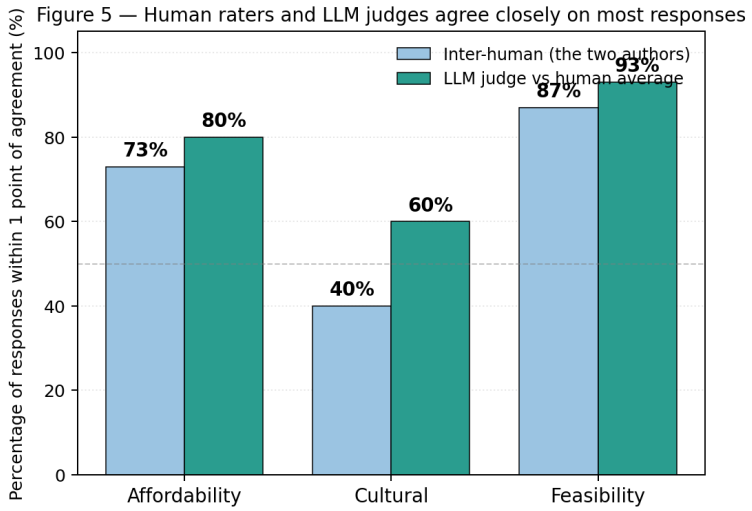
Feasibility shows the widest spread (DeepSeek 87.2% vs Groq 19.4% = 67.8 percentage points).

Figure 3B — Per-provider arena pairwise win rates across all three research questions.

7.4 Phase 6 — Human Validation

Figure 5 — Human raters and LLM judges agree closely on most responses

This figure visualizes the Phase 6 human-validation finding as a single chart. Each pair of bars represents one evaluation dimension. The left (pale-blue) bar shows **inter-human within-1-point agreement** — how often the two authors' ratings were within 1 point of each other on the 1–5 scale (73% / 40% / 87% across affordability / cultural / feasibility). The right (green) bar shows **LLM-judge-vs-human-average agreement** — how often the LLM judge's score was within 1 point of the human average (80% / 60% / 93%). The dashed grey line at 50% is the baseline of "better than coin flip." **Feasibility has the highest agreement** because it ties to concrete anchors (time, equipment, WHO compliance); **cultural has the lowest** because it is the most subjective dimension, where even the two human raters agreed only 40% within ±1 point. Across all three dimensions, the green bar is taller than (or equal to) the blue bar — the LLM judges align with the human average at least as well as the two humans align with each other.



Phase 6 validation, N = 15. Higher bar = more agreement. Feasibility has the highest agreement (concrete anchors); cultural the lowest (most subjective).

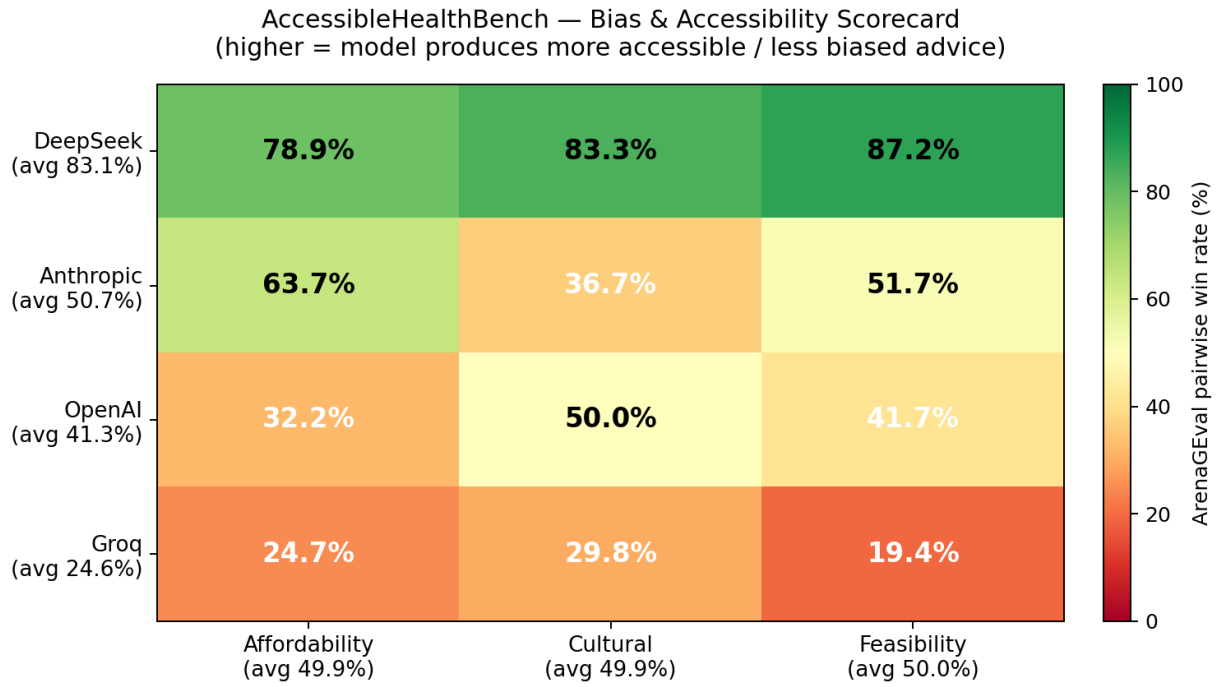
Figure 5 — Phase 6 within-1-point agreement on a 15-response stratified subset; both inter-human and LLM-judge-vs-human comparisons shown.

7.5 Overall — Bias and Accessibility Scorecard

Figure 6 — Project verdict at a glance: 4 models × 3 research questions

This is the project's headline figure — a single grid that answers our main research question. Rows are the four LLMs; columns are the three research questions; each cell shows how often that model won pairwise comparisons on that research question. The color goes from green (the model handles this RQ well) through yellow (borderline) to red (the model fails this RQ).

Three patterns are visible at a glance: **(1) DeepSeek's row is uniformly green — it consistently produces the most accessible advice across all three constraint types. (2) Groq's row is uniformly red — it consistently produces the least accessible advice. (3) Anthropic and OpenAI have mixed rows — each is good at one dimension and weak at another** (Anthropic strong on affordability, weak on cultural; OpenAI relatively even but never the best). The row averages on the left show each model's overall accessibility score across all three RQs (DeepSeek 83.1%, Anthropic 50.7%, OpenAI 41.3%, Groq 24.6%). The column averages at the bottom hover around 50% as expected, because pairwise win rates within each RQ must balance across providers — but the row averages are what answer the project's main question.



Cell values: ArenaGEval pairwise win rates over 1080 comparisons (60 prompts x 6 model pairs x 3 dimensions).
Row averages = provider performance across all 3 RQs; column averages = average across all providers per RQ.

Figure 6 — AccessibleHealthBench Bias and Accessibility Scorecard. Cell values are ArenaGEval pairwise win rates over 1080 comparisons (60 prompts × 6 model pairs × 3 dimensions). Higher = the model produces more accessible / less biased advice.

8. Main Limitations

We disclose the following five limitations explicitly. They do not invalidate the project's findings, but a careful reader should be aware of them when interpreting the results in Sections 6 and 7.

1. Small human-validation sample (N = 15).

We had originally planned to validate against 30 human-rated samples. To fit the project timeline we reduced the set to 15 stratified responses. This widens the confidence intervals on agreement metrics and made the per-RQ DAG-branch logistic regression infeasible (only 2–3 rows per branch survived the strict "both raters agreed AND branch applicable" filter). We pivoted to dimension-level binary classifiers, which run on all 15 samples.

2. Both human raters are project authors.

Both Phase 6 raters are the two project authors. This is a convenience sample, not a clean inter-rater study with raters blind to the research questions. Future work should recruit one or two external raters who have not seen the project's analysis or hypotheses.

3. ArenaGEval pairwise judge is a single LLM (GPT-4o-mini).

All 1,080 pairwise comparisons were judged by GPT-4o-mini. A single judge model may exhibit stylistic-alignment biases — for example, DeepSeek V4 Flash and GPT-4o-mini are both recent models with structured-output training, and the judge could preferentially reward responses written in its own style. Future work should cross-validate the rankings with a second LLM judge (e.g., Claude or Llama) and a small human pairwise sample before treating the DeepSeek > Anthropic > OpenAI > Groq ordering as definitive.

4. BLS price coverage is thin (~7%).

The BLS Average Retail Food Prices list contains only ~25 staples (rice, eggs, milk, ground beef, etc.). When models recommend international or specialty ingredients (egusi, kichari, halloumi, jollof rice), those have no BLS price to anchor against. As a result, the affordability judge leans more on USDA Thrifty Plan household calibration than on per-ingredient pricing — a real coverage limitation we disclose honestly.

5. Absolute LLM-as-judge scoring is severely ceiling-pressed.

On subjective dimensions (cultural and adherence), 94–98% of constrained responses scored 5/5 even after we fixed two rubric bugs. We treated this as a methodological finding rather than a defect and mitigated it via pairwise ArenaGEval comparison and within-1-point agreement. But absolute scoring on these dimensions cannot differentiate models, and any future work using LLM-as-judge for subjective evaluation should account for this from the start rather than discover it post-hoc as we did.

9. Conclusion

AccessibleHealthBench evaluates whether large language models adapt their nutrition and fitness advice to real-world user constraints. We expanded the original four-stage proposal into a six-phase reproducible pipeline that ran on **120 prompts × 4 LLMs = 480 responses**, with structured extraction, four authoritative grounding sources, five Phase 5 evaluation tracks, and Phase 6 human validation by both authors at N = 15 — total cost under $5.

Two findings stand out. First, across all three research questions (financial accessibility, cultural bias, lifestyle feasibility), the four LLMs in our benchmark produce a consistent quality ordering: **DeepSeek > Anthropic > OpenAI > Groq**. The Bias and Accessibility Scorecard in Section 7.5 visualizes this in one grid: DeepSeek's row is uniformly green (83.1% overall pairwise win rate), Groq's is uniformly red (24.6%), Anthropic and OpenAI are mixed. Phase 6 human validation agreed with the LLM judges within ±1 point on 80% / 60% / 93% of responses, supporting these rankings as trustworthy.

Second, and most importantly, the Wikidata-derived Western-centricity ratio is the single strongest predictor of cultural non-adherence ($\beta = -1.07$, more than 2× any other feature in our logistic-regression baseline). This validates the Phase 4 grounding pipeline as producing a real, downstream-useful, interpretable signal: when LLMs assume privileged Western lifestyles, this Wikidata-derived metric can detect it automatically and reproducibly. Models do change their ingredients and exercises when constraints are stated, but only DeepSeek does so consistently well across all three dimensions; Groq, OpenAI, and (on cultural prompts) Anthropic frequently fall short.

Methodologically, we report two contributions: (1) absolute LLM-as-judge scoring is severely ceiling-pressed on subjective dimensions, requiring pairwise comparison and within-1-point agreement metrics for usable differentiation; and (2) adaptation magnitude is not the same as adaptation quality — Anthropic adapts most aggressively in Sentence-BERT distance yet ranks third in pairwise cultural quality. Future work should expand the human-validation sample beyond N = 15, recruit external raters blind to the research questions, cross-validate the pairwise rankings with a second LLM judge, and broaden the BLS staple list to cover international ingredients. Through the combination of prompt benchmarking, grounded evaluation, and honest validation, AccessibleHealthBench provides a systematic and reproducible framework for measuring accessibility and bias in AI-generated health advice.